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EC39002

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Experiment 2: Design of PN Sequence Generator using LFSR



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10 February, 2026

1 Aim of the Experiment

The objectives of this experiment are to:

1. Design and implement a 5-bit Pseudo-Noise (PN) sequence generator using a Linear Feedback Shift Register (LFSR).
2. Measure and plot the auto-correlation function of a PN sequence generated using a 3-bit LFSR.
3. Investigate the "lockout" condition and implement circuitry to prevent it.

2 Components Used

- **Integrated Circuits:**
  - IC 7486 (Quad 2-input XOR gate)
  - IC 7495 (4-bit Shift Register) / IC 7474 (D Flip-Flop)
  - IC 7404 (NOT gate), IC 7408 (AND gate)
  - IC 741 (Op-Amp) for analog addition/subtraction
- **Equipment:** Digital Storage Oscilloscope (DSO), Function Generator, ±12V Power Supply.
- **Miscellaneous:** Breadboard, resistors ($1k\Omega, 10k\Omega$), connecting wires.

3 Theory

3.1 Linear Feedback Shift Register (LFSR)

A Linear Feedback Shift Register (LFSR) is a shift register whose input bit is a linear function of its previous state. The most common linear function used is the exclusive-OR (XOR).

The output sequence is deterministic but appears random, hence the name Pseudo-Noise (PN) sequence. For an N-stage register, the maximum length of the sequence before repetition is $L = 2^N - 1$.

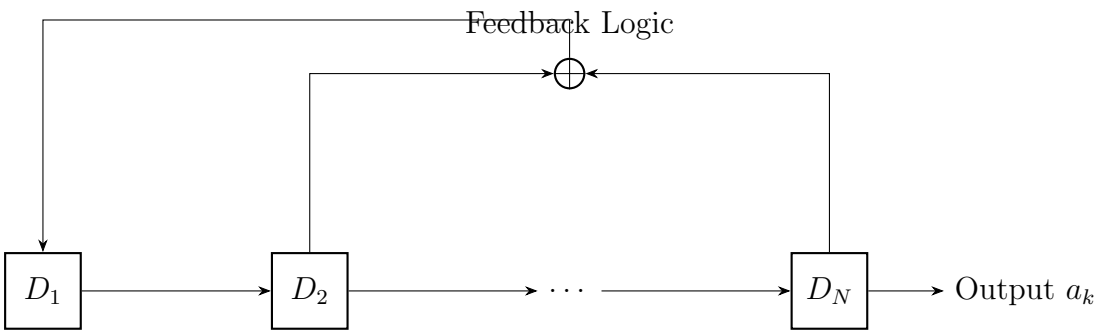


Figure 1: General Block Diagram of an N-stage LFSR

3.2 Properties of PN Sequences

1. **Balance Property:** In a maximum length sequence, the number of logic '1's is always one greater than the number of logic '0's.
2. **Run Length Property:** Half of the runs are of length 1, a quarter are of length 2, an eighth are of length 3, and so on.
3. **Auto-Correlation Property:** The auto-correlation function $R_x(k)$ is a measure of the similarity between the sequence and a phase-shifted replica of itself.

$$R_x(k) = \frac{1}{N} \sum_{i=0}^{N-1} x(i)x(i+k) \tag{1}$$

For a PN sequence, $R_x(k)$ peaks at 1 when $k = 0$ (synchronized) and drops to $-\frac{1}{N}$ for all $k \neq 0$.

3.3 Lockout Condition

The "all-zero" state is a forbidden state in a standard XOR-based LFSR. If the register enters the state $00\ldots0$, the XOR feedback ($0 \oplus 0 = 0$) will perpetually feed zeros into the input, causing a "lockout."

4 Circuit Implementation

4.1 5-bit PN Sequence Generator

A 5-bit LFSR was designed using D-flip-flops. The feedback taps were chosen to produce a maximal length sequence of $2^5 - 1 = 31$ bits.

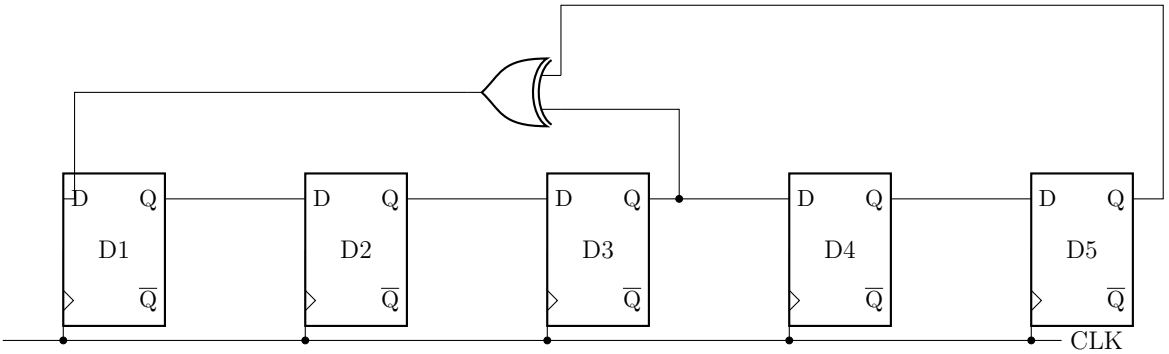


Figure 2: Circuit Schematic: 5-bit PN Sequence Generator

4.2 3-bit Auto-Correlation Circuit

To verify the correlation property, a 3-bit LFSR was used. The original sequence and a delayed version were fed into an XOR gate (acting as a multiplier), followed by an Op-Amp integration stage to extract the DC component (mean value).

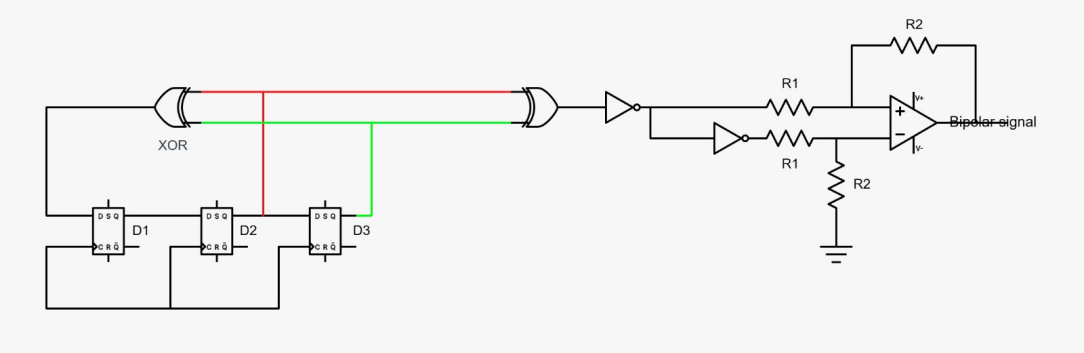


Figure 3: Circuit Schematic: 3-bit Auto-Correlation Measurement

5 Observations and Results

5.1 Generated PN Sequence

The oscilloscope capture below shows the output of the 5-bit LFSR. The sequence demonstrates the expected pseudo-random switching behavior.

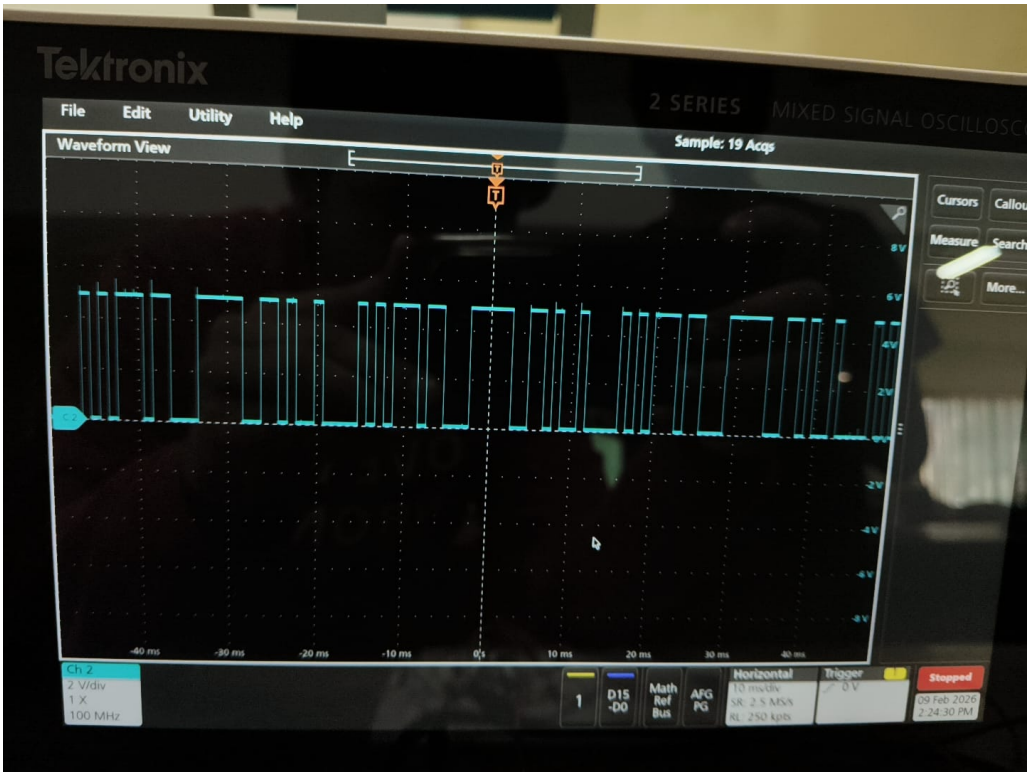


Figure 4: Oscilloscope Trace: 5-bit PN Sequence Output

5.2 Auto-Correlation Verification

The correlation property was verified by observing the output voltage of the correlator circuit under two conditions: synchronized (zero delay) and unsynchronized.

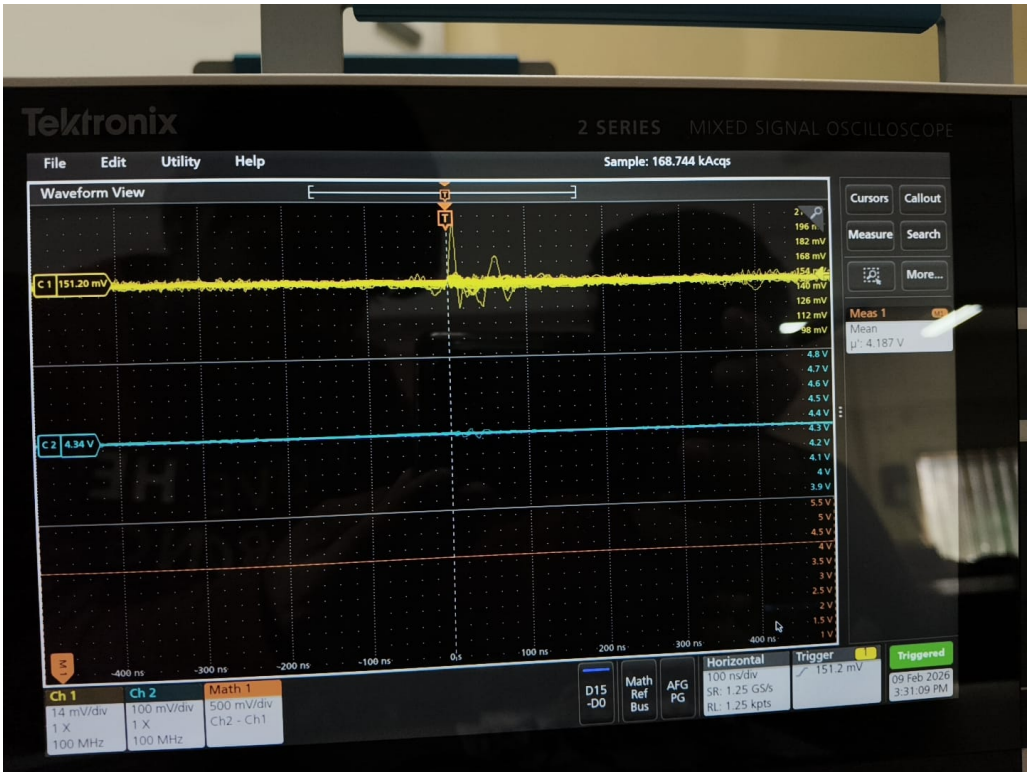


Figure 5: **Auto-correlation at zero delay ($k = 0$):** Peak correlation. Mean voltage ≈ 4.187 V.



Figure 6: **Auto-correlation at non-zero delay ($k \neq 0$):** Low correlation. Mean voltage ≈ 632 mV.

6 Discussion

In our implementation, the LFSR initially powers up with all flip-flops at logic 0. This corresponds to the forbidden all-zero state, so the XOR feedback also evaluates to 0 and the register remains stuck in the *lockout* condition. To exit lockout, we provided an initial *seed* by temporarily forcing the input of the shift register high while the clock was running. After the register state changed from all zeros, we reconnected the feedback network in its original configuration and enabled the clock again, allowing the LFSR to generate the intended maximal-length sequence.

6.1 PN Sequence Verification

Using the primitive polynomial $x^5 + x^2 + 1$, the 5-bit LFSR produced a PN sequence with period $2^5 - 1 = 31$. We verified the standard properties of an m-sequence:

- **Balance property:** In one full period we observed 16 ones and 15 zeros, i.e., the number of ones is exactly one more than the number of zeros.
- **Run-length property:** The distribution of runs (consecutive identical bits) followed the expected pattern for a maximal-length sequence, which was confirmed from the observed waveform.

6.2 Auto-correlation Verification

In the second part of the experiment, we measured the auto-correlation using the hardware correlator and estimated the mean (DC) output voltage:

- **Zero shift ($k = 0$):** Mean voltage was approximately 4.187 V (maximum correlation).
- **Non-zero shift ($k \neq 0$):** Mean voltage was approximately 0.63 V (low correlation).

This confirms the characteristic *peaky* auto-correlation property of PN sequences: a strong peak at zero shift and a much smaller value for non-zero shifts. For ideal white noise, the auto-correlation is zero for all non-zero delays; however, a PN sequence is deterministic and periodic, so its non-zero-shift correlation is not exactly zero. The observed non-zero value is therefore expected because the sequence is not truly white noise, but a periodic pseudo-random sequence.